

数学作业纸

科目_____

班级:_____

姓名:_____

编号:_____

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$$M_e = n \bar{F} \cdot \frac{D}{2} \quad \bar{F} = \frac{2Me}{n \cdot D} = \frac{2 \times 50 \cdot 10^6}{8 \times 450} \approx 27777.78 \text{ N}$$

$$\tau = \frac{F}{A} = \frac{4R}{\pi d^2} \text{ [MPa]}$$

$$d \geq \sqrt{\frac{4R}{\pi [\tau]}}$$

$$d \geq \sqrt{\frac{4 \times 27777.78}{\pi \cdot 80}} \approx 21.03 \text{ mm}$$